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AI · BAYESIAN OPTIMIZATION · US EQUITIES

Dave's Daily

tldr

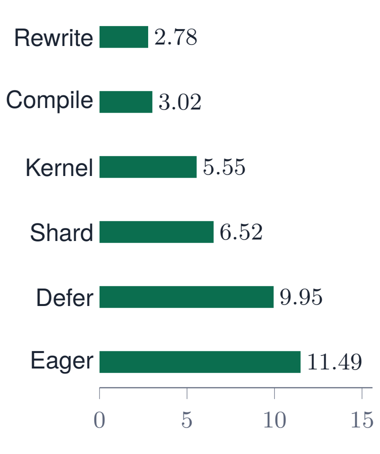
1. On Tuesday Google and Meta released TorchTPU, open-source software that runs ordinary PyTorch on Google's TPU chips. In Google's test, 4,821 of 4,917 models ran without crashing. On a FLUX training step, the same post's stack went from 11.49 seconds to 2.78 seconds.
2. Elice, a South Korean AI cloud firm, closed its first day at 60,500 won after an offer price of 90,500 won. Institutions had ordered 388.84 times the shares on offer. Reuters, citing Dealogic, called it the worst South Korean debut for a firm that raised more than $100 million.
3. STREGO, posted 9 October, is a trust-region Bayesian optimizer built for noisy scores and up to 124 inputs. At 100 dimensions, with no added noise, its mean normalized score was 0.080, against 0.354 for TURBO, on a budget of 200 evaluations.

AI

On Tuesday, at the PyTorch Conference, Google released TorchTPU as open source. Google Cloud says the stack was built with Meta's PyTorch team, and that a PyTorch program can train and serve on TPUs through the tools teams already use, including vLLM, SGLang, Hugging Face, and TorchTitan. The repository is google-pytorch/torch_tpu. TechCrunch reports that Amazon made a parallel announcement on Monday: TorchNeuron, for Amazon's Trainium chips, plus code aimed at the main PyTorch project so that features depend less on Nvidia.

The PyTorch project's own post describes the older route, PyTorch/XLA, as one that required source changes, lacked eager execution, and compiled slowly. TorchTPU treats eager mode as a requirement. A tensor moves with `.to("tpu")` in place of `.to("cuda")`. In Hugging Face Transformers v5, the same choice is `device_map="tpu"`. The post's Muse Glimmer example still replaced FlexAttention, which TorchTPU does not yet support, with a standard attention call.

The post's coverage test ran 4,917 models, on TPU generations from v5e upward. In the popularity tier, a pass means the run did not crash. That tier passed 4,821 models, reported as 98 percent. A second tier checks numerical agreement with a GPU on 657 architectures and reports 638 passing, 97 percent. On 839 causal language models, 834 matched a numerics check, 99 percent. Those counts are Google's measurements of its own software.



FLUX training step time, in seconds, after each TorchTPU software stage. Shorter is faster. The axis starts at zero. Eager 11.49; DeferAndFuse 9.95; SimpleFSDP 6.52; SplashAttention 5.55; `torch.compile` 3.02; peephole rewrites 2.78. The post does not state the chip count for this sequence. Source: PyTorch, Introducing TorchTPU, 20 October 2026.

The post gives the sample rate at the two ends of that sequence: 0.26 samples a second, then 1.08. The ratio of the step times is $11.49/2.78 = 4.13$. A separate measurement, SimpleFSDP on FLUX.1-dev, goes from one TPU v6e-8 node (8 chips) to a v6e-256 slice (256 chips), and the per-sample step moves from 0.92 seconds to 1.00 seconds. On the Google post, Meta's PyTorch engineering director, Jana van Greunen, says Meta cannot be constrained by a single hardware architecture, and that TorchTPU is already in production there. That is Meta's account, printed by Google.

Markets

The financing question showed up in a listing. Elice Group, a Seoul firm that sells AI cloud services and AI software for education and business, began trading on Tuesday. Reuters, in The Korea Times, reports that the shares closed at 60,500 won. The offer price was 90,500 won. Reuters described the decline as 32 percent and, citing Dealogic, called it the worst first day in South Korea for a company that raised more than $100 million. Elice raised 201.1 billion won, about $150 million, and said most of the proceeds would go to portable modular data centers.

Institutions had ordered 388.84 times the shares available, and the price was set on 6 October, before the recent selling in AI-related stocks. First-half revenue tripled to 34 billion won. Operating profit was 10 billion won, after a 3.5 billion won operating loss a year earlier. A profitable half and a full book still closed the first day well below the offer.

CNBC's Reuters account names three other data-center listings pulled or delayed this month: Firmus, backed by Nvidia; AirTrunk, backed by Blackstone, which postponed a Singapore REIT listing; and SB Energy, a SoftBank unit, which delayed a U.S. share sale. Ke Yan, of Shenton Research, told CNBC that the debut turns on who pays for the capacity, and for how long, until the labs that rent it can fund those commitments.

Bayesian optimization

arXiv:2610.14338, posted 9 October, is *STREGO: A Scalable Implementation of a Trust-Region Framework for Noisy Bayesian Optimization*, by Youssef Diouane, Issmail El Hallaoui, and Meisseme Kadri (GERAD and Polytechnique Montréal) and Amina Lamghari (GERAD and the Université du Québec à Trois-Rivières). The code is github.com/Mayousk/STREGO.

TREGO searches inside a trust region around the best point so far, with guarantees when every trial returns an exact value. The paper reports that those results worsen as the number of inputs grows, and exact values are rare. STREGO keeps the region and retargets the search at noisy scores in high dimension.

In the wide phase, each round picks a small batch by balancing a low predicted value against integrated variance reduction: how much a trial would cut uncertainty over the whole domain. Predictive variance, by contrast, pulls toward the boundary, where points sit far from the data. In the local phase a second Gaussian process uses only points near the current best, and the method minimizes that local prediction. Both are BoTorch `SingleTaskGP`s with one length scale per input.

Table 1, used for every experiment, sets 200 evaluations, an initial design of 10, and a global batch of 3. The synthetic suite is five functions at 4, 20, 50, and 100 dimensions (four functions at 4 dimensions, because Hartmann-6 needs at least 6). Each cell is 20 trials, with no noise, with additive noise of about 10 percent of the function's range, or with multiplicative noise of relative standard deviation 0.1. Scores are rescaled per problem so the best final value any solver reached is 0 and the worst is 1. STREGO has the lowest mean in every cell of Figures 4 to 6, between 0.061 and 0.195. At 100 dimensions with no added noise the mean is 0.080 (standard error 0.010), against 0.354 (0.036) for TURBO. Under additive noise at 4 dimensions the mean is 0.149, and the other four solvers lie between 0.474 and 0.511.

Proposal time, on an Intel Core i7-12700 using 6 cores and excluding the initial design, is 519 seconds for STREGO at 100 dimensions with no added noise, against 6,272 seconds for TREGO. At 50 dimensions under additive noise, STREGO is the cheapest of the five, at 356 seconds. These times cover the proposal step only.

On Rover (60 inputs), Mopta08 (124), and four MuJoCo tasks (Swimmer 16, Hopper 33, Walker 102, HalfCheetah 102), the paper adds no further noise. MuJoCo already perturbs the start state. Mopta08's constraints are folded in as penalties. From the curves, STREGO leads on Walker (near -800, about -490 for BAXUS) and Mopta08 (near 262, about 275 for TURBO), is slightly ahead of BAXUS on HalfCheetah, and is about even with TURBO on Swimmer. Hopper favors TURBO and BAXUS (about $-1{,}700$ and $-1{,}300$, against about $-1{,}100$). Rover favors TREGO (about 6.3, against about 6.5). Best, or tied for best, on four of the six.

Updated 21 October 2026. Phone and letter match. Sources: Google Cloud, PyTorch, and TechCrunch, 20 October; Reuters via The Korea Times and CNBC; arXiv, 9 October. Reuters calls the Elice drop 32 percent; the two prices give 33.1 percent. Dealogic is cited by Reuters. Created with malvin.